

Dylan Meeks

Date of birth: 22/03/2004 | **Nationality:** American | **Gender:** Male | **Phone number:** (+1) 9495477650 (Mobile) |
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ABOUT ME

Undergraduate student at the University of Texas at Austin studying Electrical and Computer Engineering with a focus in Computer Architecture

WORK EXPERIENCE

12/05/2024 – 20/08/2024 Irvine, United States
RESEARCH INTERN CORSA RESEARCH LAB

- Conducted exploratory research into the potential use of Dynamic Partial Reconfiguration for FPGA accelerators used in autonomous driving for lowering power requirements
- Developed proof of concept prototype that succeed in near %50 power usage drop

EDUCATION AND TRAINING

22/08/2022 – CURRENT Austin, United States
BACHELOR OF SCIENCE IN ELECTRICAL AND COMPUTER ENGINEERING The University of Texas at Austin

Website utexas.edu

22/08/2022 – CURRENT Austin, United States
MINOR IN CHINESE LANGUAGE The University of Texas at Austin

Website utexas.edu

LANGUAGE SKILLS

Mother tongue(s): **ENGLISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
CHINESE	B1	B1	B1	B1	B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

DIGITAL SKILLS

FPGA based Digital Design/Development and Testing | Arduino | LaTeX | Simulink | C++ | Verilog | HW Design with FPGA | Python | Git | Microcontrollers, Arduino, Atmega32A, PIC, STM32, FPGA, ASIC | MATLAB | C | Linux | ARM assembly | Computer architecture | Creation and assembly of circuits | Solidworks software | Xilinx Vivado/Vitis | Xilinx Vivado HLS | Virtual Memory

PROJECTS

08/2024 – 12/2024
Cycle Accurate CPU Simulator

- Developed and implemented a cycle-accurate, non-pipelined simulator for the LC-3b ISA, ensuring precision in instruction execution and clock cycle tracking.
- Designed and optimized a pipelined version of the LC-3b simulator, incorporating five-stage pipeline architecture to improve performance and throughput.
- Conducted performance analysis of pipelined and non-pipelined simulators, identifying bottlenecks and optimizing critical paths for increased efficiency.

- Employed assembly language programming to validate simulation accuracy and ensure correct operation of the LC-3b ISA features.
- Utilized advanced debugging techniques to identify and resolve issues related to pipeline hazards, data forwarding, and control flow.
- Documented design decisions, implementation details, and testing procedures for both pipelined and non-pipelined versions of the simulator.
- Applied knowledge of computer architecture principles to emulate realistic hardware behavior and accurately model LC-3b instruction set execution.
- Implemented performance optimizations, such as reducing memory usage and cycle count, to make the simulator suitable for educational use and research purposes.

11/2023 – 12/2023

FPGA Powered Digital Timer and Stopwatch

- Used HLSMs to create a digital timer/stopwatch capable of either receiving an input and counting up or down or counting up or down from fixed values, implemented using Verilog on a Digilent Basys 3 FPGA using AMD Vivado for simulations.
- Implemented using four 7-segment displays with display multiplexing to display current time, clock dividers, behavioral logic, interfacing with hardware switches, and button debouncing.

03/2023 – 04/2023

Gallaga Inspired Arcade Game

- Collaborated with team to create sprites, music, logic, and controls using C and ARM Assembly to create a rendition of the classic arcade game Gallaga.
- Designed a custom PCB using Autodesk EAGLE to interface a Ti TM4C with potentiometers, an LCD screen using the SPI protocol, a 6-bit DAC connected to both a speaker and 3.5 mm audio jack for sound.